

# Nested Limit of the Geometric Series

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Let  $n$  be a positive integer. Given that

$$a_n = \sqrt{\frac{1}{2} \sqrt{\frac{1}{4} \sqrt{\cdots \frac{1}{2^n}}}},$$

find the value of  $L$  where

$$L = \left( \lim_{n \rightarrow \infty} a_n \right)^3.$$

We have the sequence

$$a_n = \sqrt{\frac{1}{2} \sqrt{\frac{1}{4} \sqrt{\cdots \frac{1}{2^n}}}} \quad (1)$$

and we compute the first few terms in the sequence to get a sense of the general expression for what  $a_n$  will be. The  $a_1$  term is trivial and not very illustrative of the general trend, so we start with the second term

$$a_2 = \sqrt{\frac{1}{2} \sqrt{\frac{1}{4}}} = \sqrt{\frac{1}{2} \cdot \frac{1}{2}} = \frac{1}{2}. \quad (2)$$

The next term in the sequence is then

$$a_3 = \sqrt{\frac{1}{2} \sqrt{\frac{1}{4} \sqrt{\frac{1}{8}}}} = \frac{1}{2^{\frac{1}{2} + \frac{2}{4} + \frac{3}{8}}}. \quad (3)$$

A pattern is starting to emerge, but let's write down one more term to nail down what it is

$$a_4 = \sqrt{\frac{1}{2} \sqrt{\frac{1}{4} \sqrt{\frac{1}{8} \sqrt{\frac{1}{16}}}}} = 2^{-(\frac{1}{2} + \frac{2}{2^2} + \frac{3}{2^3} + \frac{4}{2^4})}. \quad (4)$$

Now it's clear what the  $n$ th term in the sequence is going to be

$$a_n = 2^{-\sum_{m=1}^n \frac{m}{2^m}}. \quad (5)$$

In the limit of  $n \rightarrow \infty$ , we can see that

$$\lim_{n \rightarrow \infty} a_n = 2^{-\sum_{n=1}^{\infty} \frac{n}{2^n}}. \quad (6)$$

In order to evaluate the sum in the exponent, recall the following series on the geometric series

$$\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n \Rightarrow \frac{x}{(1-x)^2} = \sum_{n=1}^{\infty} \frac{n}{2^n}, \quad (7)$$

and it's easier to see that the series on the right hand side actually converges to something reasonable. Letting  $x = 1/2$ , we recover the problem as stated above. The series given in the original prompt

$$\sum_{n=1}^{\infty} \frac{n}{2^n} = 2. \quad (8)$$

This implies that the original result is thus

$$\lim_{n \rightarrow \infty} a_n = 2^{-2}. \quad (9)$$

Therefore,  $L$  is simply

$$L = \frac{1}{64}. \quad (10)$$